Thermodynamics — Unit notes (Sem 4)

System and surroundings. A system is the part of the universe we study; everything else is the surroundings. An open system exchanges mass and energy, a closed system exchanges only energy, and an isolated system exchanges neither.

State variables. Pressure (P), volume (V), temperature (T) and internal energy (U) describe the state of a system. They depend only on the state, not on how the system got there. Heat (Q) and work (W) are NOT state variables — they depend on the path.

Zeroth and first law

Zeroth law. If A is in thermal equilibrium with C, and B is in thermal equilibrium with C, then A and B are in thermal equilibrium with each other. This is why a thermometer works.

First law. $\Delta U = Q - W$, where Q is heat added to the system and W is work done BY the system. It is the law of conservation of energy applied to heat and work.

Example: a gas absorbs 500 J of heat and does 200 J of work on a piston. $\Delta U = 500 - 200 = 300$ J.

Thermodynamic processes

Isothermal (constant T): for an ideal gas $\Delta U = 0$, so $Q = W$. Work = $nRT \ln(V_2/V_1)$.

Adiabatic ($Q = 0$): $\Delta U = -W$. Gas cools when it expands. $PV^{\gamma} = \text{constant}$, where $\gamma = C_p/C_v$.

Isochoric (constant V): $W = 0$, so $\Delta U = Q$.

Isobaric (constant P): $W = P\Delta V$.

On a P–V diagram, the work done is the area under the curve. An adiabatic curve is steeper than an isothermal curve through the same point.

Second law and heat engines

Kelvin–Planck statement. No engine working in a cycle can turn all the heat it takes in into work. Some heat must always be rejected to a colder sink.

Clausius statement. Heat cannot flow by itself from a colder body to a hotter body. A refrigerator needs work input to do this.

Efficiency of a heat engine: $\eta = W / Q1 = 1 - Q2/Q1$, where Q1 is heat taken from the hot source and Q2 is heat rejected to the cold sink.

Carnot engine

The Carnot cycle has four reversible steps: isothermal expansion at T1, adiabatic expansion (T1 → T2), isothermal compression at T2, adiabatic compression (T2 → T1).

Carnot efficiency: $\eta = 1 - T2/T1$ (temperatures in kelvin).

Carnot's theorem. No engine working between the same two temperatures can be more efficient than a reversible (Carnot) engine. If one could, combining it with a reversed Carnot engine would move heat from cold to hot with no work — breaking the second law. So the limit comes from the second law itself, not from friction.

Real engines are less efficient because of friction, heat leaks and irreversible processes — but even an ideal, frictionless engine cannot beat Carnot.

Example: T1 = 500 K, T2 = 300 K. $\eta = 1 - 300/500 = 0.4 = 40\%$.

Entropy

Entropy (S) measures how spread out energy is. For a reversible process, $\Delta S = Q/T$.

For an isolated system, entropy never decreases: $\Delta S \geq 0$. It stays constant only for reversible processes.

Example: 1000 J of heat flows from a 400 K body to a 200 K body. $\Delta S = -1000/400 + 1000/200 = -2.5 + 5 = +2.5$ J/K. Positive, so the process happens on its own.

Common mistake: entropy of a system CAN decrease (e.g. water freezing), as long as the entropy of the surroundings increases by more.